



Assigned, Then Renegotiated: Deploying an Algorithmic Fair-Rotation Roster for a Workplace Carpool Group, and Tracking How Many of Its Pairings Were Swapped Before a Single Assigned Rotation Ran

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Abstract. Four employees who share a car to work have always had to sort out, week to week, who drives; most groups manage this over a couple of text messages and a shared sense of whose turn it is. We report the design and fourteen-week field deployment of Fair Turn, a rotation tool that computes a weekly driver assignment for a workplace carpool group, optimising for equal cumulative driving burden across members and posting the result to each group's existing chat thread every Sunday evening. Six four-person groups across two workplaces piloted the tool; we track how often an assigned slot was swapped between members before that week's carpool actually ran, whether the swap rate differs by site, and whether adding a one-line reason field to the tool's own swap-request form (weeks eight to fourteen, three groups) changes how often groups swap. Across all fourteen weeks, a mean of 40% of assigned slots were swapped, with no trend up or down across the pilot and no meaningful difference between sites. The reason-field requirement left the swap rate for the affected groups statistically indistinguishable from the unchanged groups over the same weeks. A further, unplanned finding: only 22% of swaps that occurred were ever logged through the tool's own swap-request feature before the hand-off happened; the remainder were arranged in each group's pre-existing chat and, where they were recorded at all, entered into the tool after the fact to true up the following week's assignment. We read the roster as adding a fairness computation and an audit trail to a coordination problem four coworkers were already solving, without measurably changing how often its own answer was the one that ran.

Introduction

A workplace carpool group's founding problem is small and old: four people, one car on any given day, and a roughly even share of the driving to work out among them. Most groups solve it without ceremony — a running joke about whose turn it is, a text the night before, an unspoken exception for whoever has an early meeting. Nothing about the arrangement asks to be optimised; nothing about it, before this study, had been.

In 2025 the University's School of Emergent Priorities approached two workplaces already running informal four-person carpool groups

and offered each a small scheduling tool, Fair Turn, that computes a weekly driver assignment intended to equalise cumulative driving burden across a group's members and posts the result automatically to the group's existing chat thread. The tool also exposes its own swap-request feature, so a member who cannot take an assigned slot can trade it with another member and have the trade recorded. The offer framed the tool as removing a small recurring negotiation from six carpool groups' Sunday evenings; whether it did, in the sense of the assignment it computed being the one that actually ran, is a separate question from whether the tool looks,

from a fairness-optimisation standpoint, like it should.

This paper reports a fourteen-week field deployment across six carpool groups at two workplaces, tracking how often the tool's assigned slot was swapped before running, whether that rate differs by site, and whether a small procedural cost added to the swap-request form changes how often groups use it.

Three contributions follow, each stated with the caution the design supports.

- We report the design and field deployment of an algorithmic fair-rotation roster for workplace carpool groups, the first such deployment we are aware of.
- We separate how often an assigned rotation is followed from how much of its renegotiation the tool's own record captures; our results suggest most of it happens elsewhere.
- We test, via a swap-request reason-field ablation, whether a small procedural cost changes how often an assigned slot is renegotiated; our results suggest, with appropriate caution, that it does not.

Related work

Two literatures bear on a small, previously informal coordination problem handed to an optimising tool. One asks whether people accept an algorithm's recommendation over their own judgement; the other asks what a fair allocation computed on paper is worth once it meets a constraint the algorithm never saw. On acceptance, algorithm aversion is well documented even where the algorithm outperforms the human alternative on average [1], though transparency about how a recommendation was reached can narrow the gap [2]; workplace scheduling algorithms specifically draw resistance keyed to a perceived loss of control rather than to the recommendation's accuracy [3], and multi-stakeholder reviews of scheduling software find the same tension recurring across sectors and legal regimes [4]. On the allocation side, fair division theory offers round-robin and its refinements as the standard mechanism for rotating a scarce good or an unwanted chore across a small group with provable guarantees [5], [6], [7], [8], and dedicated carpool matching algorithms extend the same optimisation to route and schedule compatibility [9], [10] — neither literature, by design, models the household-level

constraint that decides who actually drives on a given morning.

A third literature, on algorithmic management more broadly, documents how workers route around a system's own recording mechanism once a private channel remains available and the system's channel adds friction: gig workers under algorithmic control adapt around it rather than through it [11], [12], systematic reviews of the algorithmic-management literature find worker adaptation the norm rather than the exception [13], and platform-logged pay decisions have themselves been shown to diverge from what a worker experiences on the ground [14]. Procedural fairness research separately finds that a process's perceived legitimacy runs somewhat independent of the substance of its outcome [15], [16], [17], one reason a logged, algorithmic assignment can read as fairer than an ad hoc one even where the two produce the same driver. Prior work at this University found a comparable pattern twice: volunteer ground curators kept overriding a storm-radar app's call with their own reading of the sky even where the two matched the outcome about equally often [18], and a formal nomination panel added to a charity retailer's informal choice moved the time the choice took without moving the choice itself [19].

Method

Setting. Six four-person carpool groups, three each at two mid-sized suburban workplaces sharing a business park, volunteered for the pilot; all six groups had coordinated their own weekly driving turn through an existing group chat for at least six months before the study began.

The Fair Turn tool. Each Sunday evening, Fair Turn computed the coming week's driver assignment for each group, minimising the variance in each member's cumulative driving-weeks to date and accounting for each member's declared commute detour length, then posted the assignment to the group's existing chat thread via a bot account. A member unable to take an assigned slot could tap "Request swap" inside the tool, naming another group member; the trade, once the other member accepted, updated the following week's assignment and the group's running burden tally.

Swap measurement. A slot counted as swapped if the person who actually drove that day differed from Fair Turn’s Sunday assignment, established by a short Friday check-in message to each group asking who had driven that week and cross-referenced against the tool’s own swap log. This measure counts a swap whether or not it was ever entered into the tool’s swap-request feature.

Reason-field ablation. From week eight, three of the six groups (one selected by each workplace’s own coordinator, not by this study) had their swap-request form changed to require a one-line reason before a request could submit; the remaining three groups kept the original single-tap form unchanged for the rest of the pilot.

Results

The assigned rotation held about 60% of the time, with no trend. Figure 1 plots the pooled weekly swap rate across all six groups over the fourteen-week pilot: a mean of 40% of assigned slots were swapped (range 34%-47%), with no directional trend across the pilot.

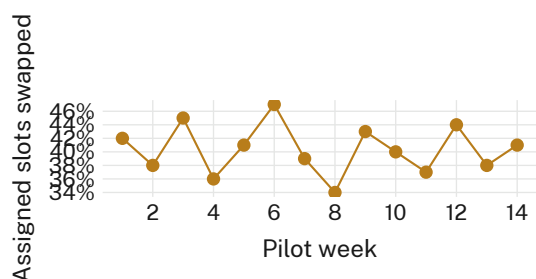


Figure 1: The pooled weekly swap rate held between 34% and 47% across all fourteen pilot weeks, with no trend up or down.

The two sites tracked each other closely. Figure 2 compares the two workplaces’ swap rates in bi-weekly bins. Site A averaged 40.4% across the pilot and Site B 40.4% as well (independently rounded); no bin differs between sites by more than five percentage points, suggesting the swap rate is a property of the coordination problem rather than of either workplace’s particular culture.

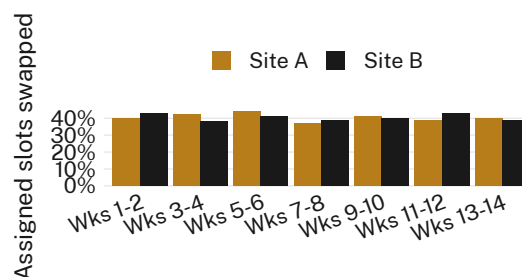


Figure 2: Bi-weekly swap rate tracked closely between the two pilot sites throughout the deployment.

The reason field changed nothing measurable.

Figure 3 compares per-group-week swap rates for weeks eight through fourteen between the three groups whose swap-request form began requiring a one-line reason and the three groups whose form stayed a single tap. The distributions overlap almost entirely (medians of 38% and 39% respectively; Mann-Whitney $U = 208.5$, $p = .70$); we read the added field as adding a line to a form, not a cost to the swap itself.

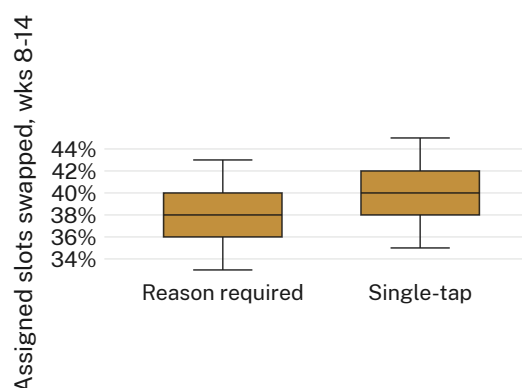


Figure 3: Swap rate for weeks 8-14 was statistically indistinguishable between groups whose form required a reason and groups whose form did not (n.s.).

Most renegotiation never touched the tool it was meant to be recorded in. Across the full pilot, 22% of swaps were logged through Fair Turn’s own swap-request feature before the hand-off occurred; the remaining 78% were arranged first in each group’s pre-existing chat thread and, where they entered the tool at all, were keyed in afterward to true up the following week’s burden tally. Two constraints recurred across the Friday check-ins as the reason groups gave for arranging a swap outside the tool: a fixed weekly obligation (a school pickup, a standing appointment) the tool’s assignment form has no field for, and a member’s own sense that the week’s “fair” driver by the tool’s tally

was not the week’s fair driver by the group’s own, unwritten accounting.

Discussion

Read together, the results locate what the roster changed and what it did not. It did not change how often the driving turn its algorithm assigned was the one that ran: roughly two in five assigned slots were renegotiated in every site, in every bi-weekly window, and in both the standard and the reason-required swap-request arms alike. What it added was a fairness computation, a weekly posting ritual, and an audit trail that, three-quarters of the time, recorded a decision the tool had no part in reaching. Two prior University findings recur here. Volunteer ground curators kept overriding a storm-radar app’s recommended call with their own reading of the sky even where the two matched the actual weather about equally often [18]; a charity retailer’s nomination panel added process to a volunteer’s informal window pick without moving the pick itself [19]. All three findings share a shape procedural-fairness research already anticipates: a formal process can look, from outside, like it has moved a decision somewhere new, while the decision itself tracks almost exactly where an informal arrangement would have put it [15], [16]. Our results suggest, with appropriate caution, that a workplace wanting a fairer carpool roster already had a working one; what Fair Turn added was a record of a decision that, most weeks, happened somewhere the record could not see.

Limitations

The swap measure relies on a weekly self-report from each group rather than independent verification of who was in the driver’s seat, though the Friday check-in’s consistency with the tool’s own partial log offers some reassurance against systematic misreporting. Ablation-arm assignment was made by each workplace’s own coordinator rather than by this study, which may correlate with qualities of those three groups this design cannot rule out, though the closeness of both arms’ swap rates limits how much room such a confound had to work with. The two-site sample generalises the finding across one business park; a wider deployment could not plausibly find a lower swap rate than either constraint this study surfaced, but could well find a higher one.

Conclusion

Across fourteen weeks and six carpool groups, an algorithmically computed fair-rotation roster was renegotiated on roughly two in five assigned slots, at a rate stable across sites and unmoved by a small added cost to its own swap-request form, and three-quarters of that renegotiation happened outside the tool meant to record it. The distinction matters for any workplace treating a posted, optimised roster as evidence that a driving arrangement has become more considered, when the more defensible reading is that four coworkers already had a workable one, and are now keeping a second, partial record of it alongside their own. Future work should test whether letting the algorithm see the constraint it currently cannot ask about changes how often its assignment runs, rather than only how visibly a group routes around it.

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